

Subhradeep Acharjee

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EDUCATION

New York University

Sep 2025 - May 2027

Masters, *Computer Science* | *GenAI, ML, RL/LM & Reasoning, Database, Computer Vision, Algorithms, Big Data*

BMS Institute of Technology and Management

Aug 2019 - Jun 2023

B.E., *Computer Science*

SKILLS

Technical Skills: Languages: SQL, Python, JavaScript, Shell Script, HTML, Backend & Data: FastAPI, Data Modeling, Node.js, Express.js, MongoDB, PostgreSQL, PostGIS, BigQuery, Apache Spark, Hive, Supabase, AI/ML: PyTorch, CUDA, NVIDIA NeMo, llama-cpp, HuggingFace, Vertex AI, Google ADK, RAG, LLM, Cloud & Tools: Pytest, MyPy, OAuth2, bcrypt, Cloud Run, Git, ServiceNow, Weights & Biases, Figma, Linux

Soft Skills: Communication & Collaboration, Problem-Solving & Critical Thinking, Adaptability & Learning Agility, Project Management

WORK EXPERIENCE

Prodapt Solutions | Software Engineer

Jan 2024 - Jul 2025

- Automated JIRA ticket generation and a router validation framework with real-time config fetching, pre/post-upgrade verification, and on-demand scheduling - reducing manual intervention by 30%, improving network reliability by 25%, and enabling continuous monitoring across 500+ devices.
- Designed optimized database schema for network data retrieval and updates, reducing query time by 40%; integrated REST APIs for workflow orchestration across process, network, and schema layers.
- Developed ServiceNow client scripts, business rules, SLAs, notifications, and UI policies to enhance platform automation, user experience, and service compliance.

Varcons Technologies | Web developer

Feb 2023 - Apr 2023

- Built and deployed a responsive website for LS Cafe; designed an interactive figma prototype for UI/UX planning, streamlining the development workflow and implemented an email subscription system to boost customer engagement and retention.

PROJECTS

RiskTwin: Multimodal Industrial Safety Sentinel |

Aug 2026 -

- Engineered a local multimodal safety system for crane lift operations using 8 specialist AI agents (vision, ASR, telemetry, policy gate) with COLMAP 3D reconstruction to flag unsafe spatial configurations before each lift proceeds.
- Containerized full pipeline with Docker Compose, MongoDB persistence, and dual hardware profiles (Mac dev, NVIDIA DGX GB10 production); automated test suites covering all 8 agent decision branches.

howToTrainYourRobo: Physical Data Marketplace for Robot Learning |

June 2026 -

- Built an end-to-end robotics data pipeline - iOS ARKit/LiDAR capture → Modal GPU cloud → 3D Gaussian Splatting reconstruction + MediaPipe pose extraction + YOLO object detection - producing structured training data for robot learning models.
- Designed a Gemini Vision evaluation layer to automatically score data quality across captures, reducing manual annotation effort and enabling scalable ground-truth generation for embodied AI datasets.

SMB Underwriting |

Apr 2026 -

- Built a fully automated ML pipeline for SMB credit risk assessment - credit scoring model, default probability forecasting, and causal "what-if" queries enabling loan officers to simulate counterfactual approval scenarios.
- Engineered feature preprocessing, model selection, and a causal inference layer in Python that produced interpretable risk scores with per-feature attribution, making model decisions auditable and explainable to non-technical stakeholders.

LM_SVG: Transformer Scaling Laws on SVG Code |

Feb 2026 -

- Trained decoder-only Transformer LMs at 5 scales (1M-88M params) on 123M SVG tokens; fit power-law scaling curves achieving exponent $\alpha = 0.2206$ (standard) and $\alpha = 0.2360$ with muP, a 6.9% efficiency gain.
- Implemented Maximal Update Parameterization (muP) enabling zero-shot learning rate transfer across all model widths without divergence, eliminating per-scale hyperparameter tuning.
- Achieved test perplexity of 2.03 and 23.5% zero-shot valid render rate on the 88M XL model, demonstrating emergent geometric continuity and SVG syntax understanding.

COCURRICULUR & ACHIEVEMENTS

- CA @ NYU - Mentoring 30 master students
- currently enrolled in CodePath Applied AI Engineering Program
- Prodapt University Recognition: Mastered essential professional skills for enhanced career readiness.
- Hackathon wins - Nvidia spark 2026, ML X Finance(Intuit), AI healthcare, AI factory(Dell X Nvidia)